Yeast Cell Populations producing Toxic Products in Fed-Batch Bioreactors

Elias Riedling, Kaize Zhou

Computational Ecology II - SS 2026

1 Introduction

Fossil fuels are a primary driver of global warming and environmental pollution, yet fossil fuels still account for approximately 77% of global energy consumption [7]. In response, alternative energy sources have been developed. One such energy source is bioethanol, which can help to reduce the total carbon footprint by partly replacing fossil fuels. [3] In many processes, *Saccharomyces cerevisiae* is used for the production of bioethanol. [5]
S. cerevisiae exhibits fast growth rates and is relatively easy to genetically manipulate. [2] Therefore, yeast cells are extensively used in biotechnological applications, including the production of bioethanol. For industrial biotechnological processes containing yeast, fed-batch processes are commonly used. [2] A fed-batch culture is one of the most widely used cultivation strategies in industrial biotechnology, in which nutrients are added to the bioreactor during cultivation, while the product is typically harvested at the end of the cultivation process. One major benefit of this cultivation strategy is that a low sugar concentration is maintained throughout the process, which results in an efficient conversion of fermentable sugars into ethanol. By adjusting the rate of nutrient supply, the cellular growth rate can be regulated. In our model, this allows the yeast growth rate to be treated as a controllable parameter. [5, 8]

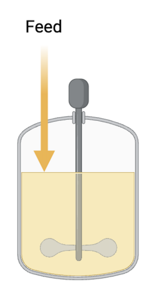

Figure 1: Typical Fed-Batch reactor setup used in biotechnological processes. The reactor is equipped with a stirrer to homogenize the solution. Nutrients are continuously supplied through a controlled feed.

However, challenges arise when the product is toxic to the host cells. This is also the case for the production of bioethanol, as ethanol has a toxic effect on yeast cells [1]. Since product removal is not considered in our modeled fed-batch system, chemical degradation is the only mechanism by which the product concentration can decrease. However, the degradation of ethanol occurs relatively slowly under certain bioreactor conditions [4]. Although certain yeast strains can exhibit relatively high ethanol productivity and ethanol tolerance, increasing ethanol concentrations still eventually inhibit yeast growth and reduce cellular viability and, consequently, ethanol production. Therefore, the accumulation of ethanol represents an important limitation for achieving high product concentrations in fed-batch cultivation [5].

Although ethanol toxicity in yeast is well established, mathematical models explicitly describing the feedback between a yeast population and a toxic metabolic product remain limited. Therefore, this research project aims to model the dynamics of a yeast cell population in the presence of a toxic metabolic product and how this interaction affects the stability and equilibrium of the population. A secondary aim of this research project is to maximize the achievable product concentration while minimizing the time required to reach it.

2 The model

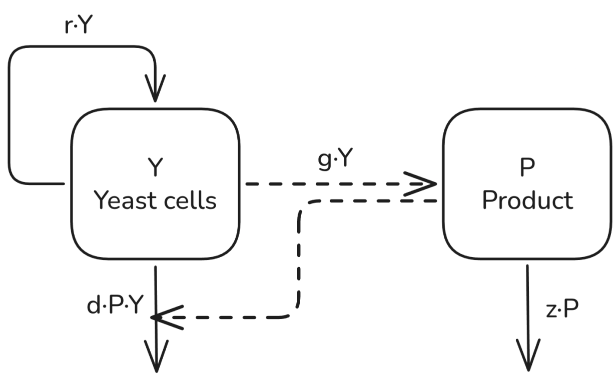

Figure 2: Flow chart of the model describing yeast growth and the formation of a toxic product. Yeast cells Y grow at rate rY and produce product P at rate gY; product decays at a rate zP and increases yeast mortality at rate dPY. Arrows indicate the direction of each flow between Y and P.

The model developed in this research project describes a yeast cell population $Y(t)$ growing in a fed-batch reactor subject to a continuous supply of nutrients. The yeast cells produce a product whose concentration is represented by $P(t)$ over time, which exerts a toxic effect on the yeast cell population. In this model, $Y(t)$ and $P(t)$ are tracked continuously over time.

By adjusting the rate of nutrient supply, the growth rate of the yeast population can be controlled. In our model, we assume that the rate of nutrient supply is adjusted such that the yeast cell population grows at a rate of rY, where $0 < r < 1$ is the specific growth rate and Y is the yeast cell population. The specific product generation rate is represented by $0 < g < 1$ and is generated in proportion to the population size. Product decay is modeled as zP, where $0 < z < 1$ is the product decay rate and P is the product concentration. Additionally, the product concentration influences the death term of the yeast cells, which is modeled as dPY, where $0 < d < 1$ is the specific death rate of yeast cells, P the product concentration and Y the yeast cell population.

The resulting model equations:

$$\frac{dY}{dt} = rY - dPY \tag{1}$$

$$\frac{dP}{dt} = gY - zP \tag{2}$$

Thus, the model describes the following sequence of events: nutrient supply promotes yeast growth and the growing yeast population produces more product. This product accumulates and decays over time, and increasing product levels increase yeast mortality. These processes determine the changes in Y and P over time.

3 Results

3.1 Dynamics of the yeast population and product concentration

Starting from a small initial yeast population, the simulation showed that the yeast population and product concentration do not increase indefinitely but instead converge toward a stable equilibrium. Initially, when P and Y are low, the growth of the yeast population is primarily determined by the intrinsic growth term rY. However, an increase in the yeast population Y leads to an increase in product concentration P. The resulting increase in product concentration subsequently reduces the yeast population Y due to the toxic effect of the product P. This interaction can be observed in Figure 3, where both Y and P converge toward stable equilibrium values over time, reaching 1.2 for the product concentration and 0.6 for the yeast population, respectively.

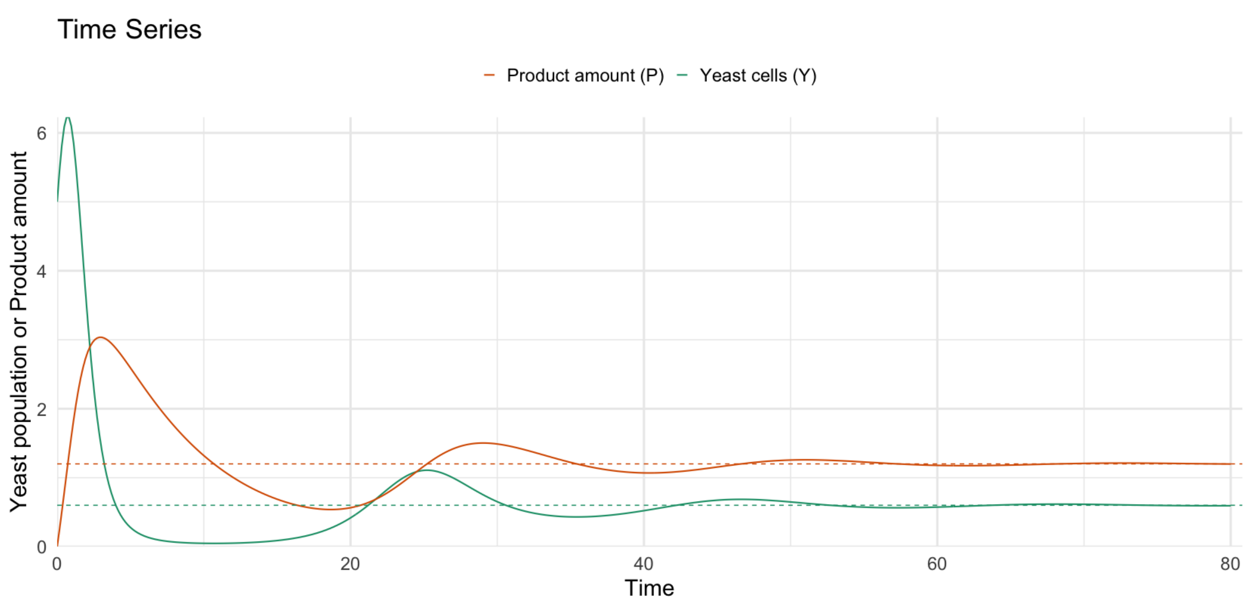

Figure 3: Yeast cell population and product concentration over time with the following parameters: $r = 0.6$, $g = 0.3$, $d = 0.5$, $z = 0.15$ and a starting population of $Y(0) = 5$. The green line represents the yeast population, while the orange line represents the product concentration.

3.2 Effect of parameters on the equilibrium

The model has a trivial equilibrium at $Y = 0$ and $P = 0$, corresponding to a yeast population that has gone extinct and, consequently, no toxic product can be generated. The nontrivial equilibrium values for the yeast cell population and product concentration were derived as follows:

$$\hat{Y} = \frac{rz}{dg} \qquad \hat{P} = \frac{r}{d}$$

where

r : replication rate, z : product decay rate,

d : specific death rate of the yeast cells, g : product generation rate.

Interestingly, the equilibrium product concentration is dependent only on the specific replication rate r and death rate d of the yeast population. The product generation rate g and the product decay rate z do not affect the equilibrium product concentration. In comparison with Figure 3, Figure 4 shows that increasing the replication rate r and decreasing the death rate d lead to an increase in the equilibrium product concentration (left), whereas variations in g and z alone do not affect the product equilibrium (right).
In contrast, the equilibrium of the yeast cell population is directly proportional to both the replication rate r and the product decay rate z. Increasing the specific death rate of the yeast cells d and product generation rate g, however, results in a lower equilibrium yeast population. Thus, while r and d determine the equilibrium product concentration, the equilibrium yeast population is affected by all four parameters. In comparison to Figure 3, Figure 4 also shows that changes in the parameters z, r, g or d lead to a different yeast population equilibrium.

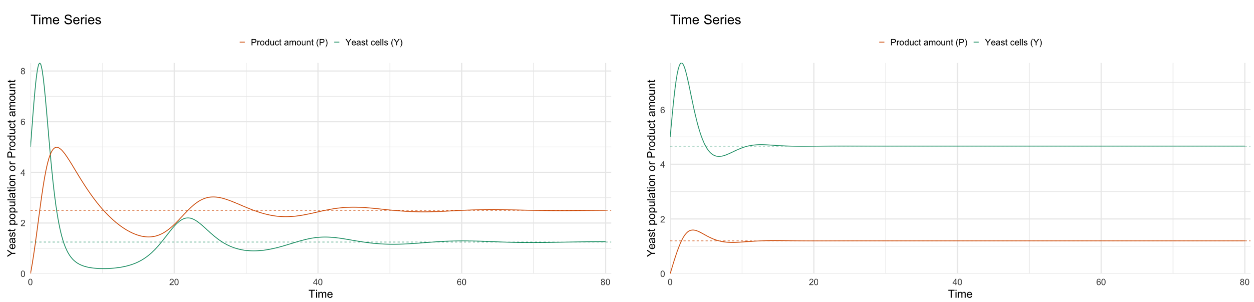

Figure 4: Yeast cell population and product concentration over time for two different parameter sets, both with an initial yeast population of $Y(0) = 5$. Left: $r = 0.75$, $g = 0.3$, $d = 0.2$, and $z = 0.15$. Right: $r = 0.6$, $g = 0.18$, $d = 0.5$, and $z = 0.7$. The green line represents the yeast population, while the orange line represents the product concentration.

3.3 Stability analysis of the equilibria

The stability analysis showed that the nontrivial equilibrium is stable and non oscillatory when $z \geq 4r$. In this case, both eigenvalues are negative, resulting in a stable and non-oscillatory equilibrium. In contrast, for $z < 4r$, the eigenvalues form a complex conjugate pair with real negative parts. Consequently, the system has a stable equilibrium that oscillates with a decaying amplitude as the system converges toward the equilibrium.

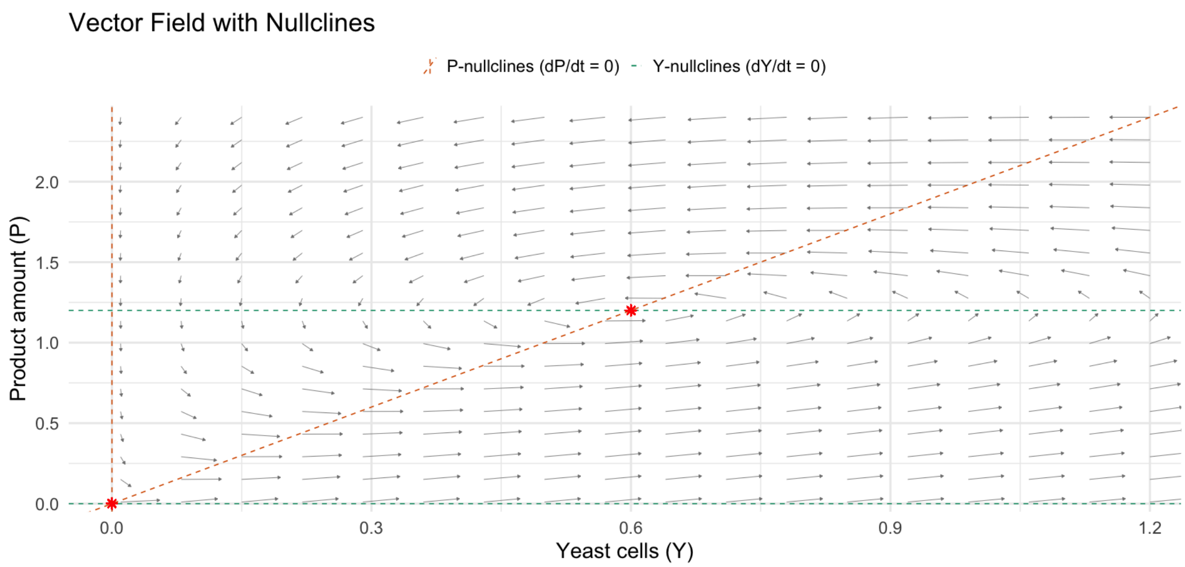

Figure 5: Vector field analysis with the following parameters: $r = 0.6$, $g = 0.3$, $d = 0.5$, $z = 0.15$ and a starting population of $Y(0) = 5$. The orange stars represent equilibria and the arrows indicate the dynamics in the plane.

4 Discussion

The model demonstrates that the interaction between the growth of the yeast population and the accumulation of toxic product results in a stable equilibrium for both the yeast population and the product concentration. These results indicate that the product toxicity represents an important limiting factor for the total achievable product concentration in the modeled fermentation system.
A key finding of this model is that the equilibrium product concentration is determined only by the replication rate r and the specific death rate d, as described by $\hat{P} = r/d$. Therefore, the product generation rate g and product decay rate z do not directly affect the equilibrium product concentration. This suggests that one could optimize product yield by maximizing the replication rate r and minimizing the death rate d, but since these parameters reflect fixed biological properties of the yeast population, they cannot necessarily be adjusted independently in a real production process. Another key finding is that the product decay rate z and product generation rate g nevertheless influence the equilibrium yeast population according to $\hat{Y} = rz/(dg)$. This can be explained by the influence of these factors on the accumulation of toxic product.
The stability analysis demonstrates that the nontrivial equilibrium is stable for positive parameter values. For $z \geq 4r$ the system approaches the equilibrium without oscillations, whereas for $z < 4r$, the equilibrium is approached through oscillations. These oscillations can be biologically interpreted as a consequence of the interaction between yeast growth and product toxicity, which causes temporary oscillations in both variables before converging to their respective equilibrium.
Despite providing insights into the interaction between yeast growth and product accumulation, the model represents a simple description of the interaction and does not capture several biological processes that may occur in real fermentation systems. In particular, parameters r, g, d and z are assumed to remain constant over time. These assumptions represent a limitation of the model, as factors such as the yeast physiology are not explicitly taken into account. Consequently, some predictions of the model may not be directly translatable into real bioethanol production processes.
The model suggests that, to maximize the achievable product concentration, the product should

be removed at the first maximum of P, before product toxicity suppresses the yeast population. However, this assumption is biologically unrealistic, since the product generation rate g is not necessarily constant over time. Instead, the product formation also depends on the physiological state of the yeast population, which may strongly vary throughout the bioethanol formation process which may result in low product formation rates, particularly during the initial phase of fermentation. [6] Consequently, the predicted equilibria should be regarded as theoretical outcomes of the model rather than directly predicting the industrial bioethanol production process.

Future models could incorporate these dynamic parameters. Such extensions could enable more realistic predictions of the dynamics and potential optimizations of toxic-product processes, including the production of bioethanol.

5 Appendix

5.1 Mathematical analysis

Model equations

$$\frac{dY}{dt} = rY - dPY \tag{3}$$
$$\frac{dP}{dt} = gY - zP \tag{4}$$

Equilibria

Setting $\frac{dP}{dt} = \hat{P} = 0$:

$$g\hat{Y} - z\hat{P} = 0 \tag{5}$$
$$\hat{P} = \frac{g}{z}\hat{Y} \tag{6}$$

Setting $\frac{dY}{dt} = \hat{Y} = 0$:

$$0 = r\hat{Y} - d\hat{P}\hat{Y} \tag{7}$$
$$\Rightarrow \hat{Y} = 0 \quad \text{or} \quad \hat{P} = \frac{r}{d} \tag{8}$$

Combining $\hat{P} = r/d$ with $\hat{P} = (g/z)\hat{Y}$ gives the nontrivial equilibrium:

$$0 = r - d\hat{P} \tag{9}$$
$$0 = r - \frac{dg}{z}\hat{Y} \tag{10}$$
$$\boxed{\hat{Y} = \frac{rz}{dg}} \qquad \boxed{\hat{P} = \frac{r}{d}} \tag{11}$$

Jacobian matrix

$$J = \begin{pmatrix} \frac{\partial Y}{\partial Y} & \frac{\partial Y}{\partial P} \\ \frac{\partial P}{\partial Y} & \frac{\partial P}{\partial P} \end{pmatrix} = \begin{pmatrix} r - d\hat{P} & -d\hat{Y} \\ g & -z \end{pmatrix}$$

Evaluated at the nontrivial equilibrium $(\hat{Y}, \hat{P}) = \left(\frac{rz}{dg}, \frac{r}{d}\right)$:

$$J\Big|_{\hat{Y},\hat{P}} = \begin{pmatrix} 0 & -\frac{rz}{g} \\ g & -z \end{pmatrix}$$

Eigenvalues

$$\lambda_{1,2} = \frac{-gz \pm \sqrt{g^2 z(z-4r)}}{2g} \overset{g>0}{=} \frac{-z \pm \sqrt{z(z-4r)}}{2}$$

$$\lambda_1 = \frac{-z - \sqrt{z(z-4r)}}{2}, \qquad \lambda_2 = \frac{-z + \sqrt{z(z-4r)}}{2}$$

Oscillations

Oscillations are possible if the discriminant is negative:

$$z(z-4r) < 0 \iff z^2 - 4rz < 0 \iff z - 4r < 0 \iff \boxed{\frac{z}{r} < 4}$$

If oscillations occur, the equilibrium is stable, since

$$\mathrm{Re}(\lambda_{1,2}) = -\frac{z}{2} < 0 \forall z.$$

No-oscillation case ($z \geq 4r$)

$$z(z-4r) \geq 0 \ \Rightarrow \ z - 4r \geq 0 \ \Rightarrow \ z \geq 4r$$

(since $4r > 0$, we always have $z - 4r \leq z$).

Stability of λ_2:

$$\begin{aligned}
\frac{-z + \sqrt{z(z-4r)}}{2} &< 0 && (12)\\
\sqrt{z(z-4r)} &< z && (13)\\
z(z-4r) &< z^2 \qquad \text{(squaring, valid since both sides} > 0) && (14)\\
z^2 - 4zr &< z^2 && (15)\\
-4zr &< 0 \qquad \forall\, z, r > 0 \quad \text{(always true)} && (16)
\end{aligned}$$

Stability of λ_1:

$$\begin{aligned}
\frac{-z - \sqrt{z(z-4r)}}{2} &< 0 && (17)\\
-z - \sqrt{z(z-4r)} &< 0 && (18)
\end{aligned}$$

Since $z > 0$ and $\sqrt{z(z-4r)} \geq 0$ for $z \geq 4r$, this is always negative.

Conclusion

The equilibrium is **always stable**, and **oscillation-free for** $z \geq 4r$.

Simulations

Link to our shiny-app for visualisations and simulations:
https://lai-es-toxic-yeast-production.share.connect.posit.cloud/

References

[1] Elisabetta Canetta, Ashok K. Adya, and Graeme M. Walker. "Atomic force microscopic study of the effects of ethanol on yeast cell surface morphology". In: *FEMS Microbiology Letters* 255.2 (Feb. 2006), pp. 308–315. ISSN: 0378-1097. DOI: 10.1111/j.1574-6968.2005.00089.x. eprint: https://academic.oup.com/femsle/article-pdf/255/2/308/19119622/255-2-308.pdf. URL: https://doi.org/10.1111/j.1574-6968.2005.00089.x.

[2] Geoffrey P Lin Cereghino and James M Cregg. "Applications of yeast in biotechnology: protein production and genetic analysis". In: *Current Opinion in Biotechnology* 10.5 (1999), pp. 422–427. ISSN: 0958-1669. DOI: https://doi.org/10.1016/S0958-1669(99)00004-X. URL: https://www.sciencedirect.com/science/article/pii/S095816699900004X.

[3] Ayhan Demirbas. "Biofuels securing the planet's future energy needs". In: *Energy conversion and management* 50.9 (2009), pp. 2239–2249.

[4] P. Duboc and U. von Stockar. "Systematic errors in data evaluation due to ethanol stripping and water vaporization". In: *Biotechnology and Bioengineering* 58.4 (1998), pp. 428–439.

[5] Siti Hajar Mohd Azhar et al. "Yeasts in sustainable bioethanol production: A review". In: *Biochemistry and Biophysics Reports* 10 (2017), pp. 52–61. ISSN: 2405-5808. DOI: https://doi.org/10.1016/j.bbrep.2017.03.003. URL: https://www.sciencedirect.com/science/article/pii/S2405580816302424.

[6] Danilo Porro, Luca Brambilla, and Lilia Alberghina. "Glucose metabolism and cell size in continuous cultures of Saccharomyces cerevisiae". In: *FEMS Microbiology Letters* 229.2 (Dec. 2003), pp. 165–171. ISSN: 0378-1097. DOI: 10.1016/S0378-1097(03)00815-2. URL: https://doi.org/10.1016/S0378-1097(03)00815-2.

[7] Hannah Ritchie and Pablo Rosado. "Energy Mix". In: *Our World in Data* (2020). URL: https://ourworldindata.org/energy-mix.

[8] Tsuneo Yamanè and Shoichi Shimizu. "Fed-batch techniques in microbial processes". In: *Bioprocess Parameter Control*. Berlin, Heidelberg: Springer Berlin Heidelberg, 1984, pp. 147–194. ISBN: 978-3-540-39004-6.

Documentation of the use of Generative AI Systems

In the overview below, I fully document the use of generative AI systems in the preparation of this written work and/or the underlying research and analyses.

I hereby declare [please select the applicable option] …

- o … that no generative AI was used for this work and that I prepared and wrote it entirely on my own. Accordingly, the following documentation is not applicable.
- X … that I have fully documented the type and scope of the use of generative AI applications as auxiliary tools in the table below, and that my own creative contribution predominates in the submitted written work[1]

Item [Existing entries = examples]	**Generative AI system(s) used**	**Detailed description of the AI usage**
Concept development: Focal topics of the work.		
Literature work: Research and content analysis		
Data analysis: Selection of procedures or interpretation		
Data processing: Creating or revising program code (e.g., R)	Claude	Initializing the basics of The Shiny-app in R
Creation of Figures or Tables.		
Writing: Text structure		
Writing: Language editing	ChatGPT	ChatGPT corrected for potential language mistakes and content mistakes in our Draft
Writing: Translations		
Please add any additional work steps not listed here.		
Please add any additional work steps not listed here.		
Please add any additional work steps not listed here.		

Place, Date, Signature

Tübingen, 13.08.2026

[1] The list of keywords in the template is neither exhaustive nor mandatory to follow. Use it as a source of inspiration. Document your specific AI use in a meaningful and transparent way. If you are unsure whether you need to declare the use of a particular AI system, please contact your supervisor. [Please remove this footnote]